

AIM:

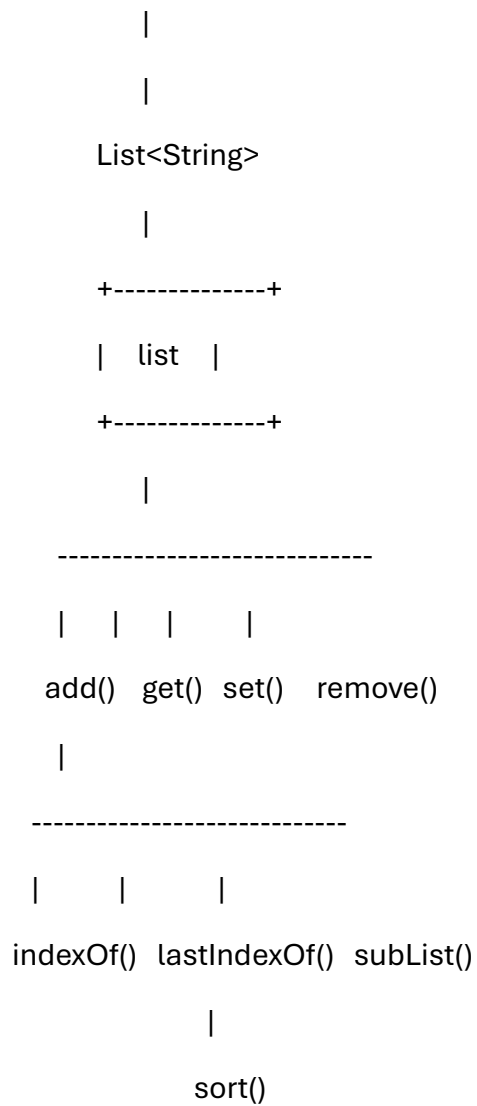
To write, compile, and execute a Java program to demonstrate the commonly used methods of the List interface, including add(), get(), set(), remove(), indexOf(), lastIndexOf(), subList(), and sort().

ALGORITHM

1. **Start** the program.
2. Import the java.util.* package.
3. Create a List<String> using the ArrayList class.
4. Use add(E e) to append elements to the list.
5. Use add(int index, E element) to insert an element at a specified position.
6. Use get(int index) to retrieve an element from a specified index.
7. Use set(int index, E element) to replace an existing element.
8. Use remove(int index) to remove an element from the specified index.
9. Use indexOf(Object o) to find the first occurrence of an element.
10. Use lastIndexOf(Object o) to find the last occurrence of an element.
11. Use subList() to obtain a portion of the list.
12. Use sort() to arrange the elements in ascending order.
13. Display the results of all operations.
14. **Stop** the program.

UML Diagram:

```
<<interface>>
List<E>
  △
  |
  implements
  |
+-----+
| ArrayList<E> |
+-----+
```



Code:

```

package Observations;

import java.util.*;

public class ListInterface {

    public static void main(String[] args) {

        List<String> list = new ArrayList<>();

        // add(E e)

```

```
list.add("Apple");  
list.add("Mango");  
list.add("Banana");  
list.add("Mango");
```

```
System.out.println("Original List: " + list)
```

```
list.add(1, "Orange");  
System.out.println("After adding at index 1: " + list);
```

```
// get(int index)  
System.out.println("Element at index 2: " + list.get(2));
```

```
// set(int index, E element)  
list.set(2, "Grapes");  
System.out.println("After setting index 2: " + list);
```

```
// remove(int index)  
list.remove(3);  
System.out.println("After removing index 3: " + list);
```

```
// indexOf(Object o)  
System.out.println("First index of Mango: "  
    + list.indexOf("Mango"));
```

```
// lastIndexOf(Object o)  
System.out.println("Last index of Mango: "  
    + list.lastIndexOf("Mango"));
```

```

// subList(int fromIndex, int toIndex)

List<String> sub = list.subList(1, 3);

System.out.println("SubList: " + sub);


// sort(Comparator)

list.sort(Comparator.naturalOrder());

System.out.println("Sorted List: " + list);

}

}

```

Output:

```

'bservations.ListInterface'
Original List: [Apple, Mango, Banana, Mango]
After adding at index 1: [Apple, Orange, Mango, Banana, Mango]
Element at index 2: Mango
After setting index 2: [Apple, Orange, Grapes, Banana, Mango]
After removing index 3: [Apple, Orange, Grapes, Mango]
First index of Mango: 3
Last index of Mango: 3
SubList: [Orange, Grapes]
Sorted List: [Apple, Grapes, Mango, Orange]
PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java>

```

RESULT:

Thus, the Java program demonstrating the various methods of the List interface was successfully implemented, compiled, and executed, and all the specified list operations were performed successfully.